



SCHOOL OF COMPUTING & ENGINEERING SCIENCES

BACHELOR OF SCIENCE IN COMPUTER NETWORKS AND SECURITY

COURSE OUTLINE

Course Code/Name	CNS 2201 Cryptography I
Lecturer	
Lecturer Contact / Office	T Lonyeiyi / Corporate Staffroom, door no. ADIF 10
Course Link	
Module Leader	

1. Purpose of the Course

This course introduces students to basic concepts in cryptography and discusses both their theoretical foundations and practical applications.

2. Expected Learning Outcomes

At the end of this course, the student will be able to:

1. Analyze various issues related to computer security.
2. Compare the features provided by public-key and private-key encryption systems.
3. Review the strengths and weaknesses of current data encryption algorithms.

3. Prerequisites

- CNS 2103– Introduction to Cyber Security
- CNS 2102 – Data structures and Algorithms

4. Detailed Course Outline/Content

Week	Topic	Discussion Topics	Activities
1	Introduction	• Overview of cryptology • Overview Cryptographic Techniques	• Class discussions • Class exercises
2	Stream Ciphers	• Stream vs Block Ciphers • Random Numbers and an Unbreakable Stream Cipher • Shift Register-Based Stream Cipher	• Class discussions • Class exercises
3	Block Ciphers and Data Encryption Standard (DES)	• Introduction to DES • Internal Structure of DES • Security of DES • DES alternatives • Modes of operation of block ciphers: ECB, CBS, OFB, CFB, CTR, GCM	• Class discussions • Class exercises • Lab exercise
3-4	Advanced Encryption	• Mathematics: Galois Fields • Internal structure of AES	• Class discussions • Class exercises

	Standard (AES)	• Decryption • Implementation in software and hardware	
5	CAT 1		
5-6	Introduction to Public Key Cryptography	• Symmetric vs Asymmetric Cryptography • Practical aspects of public key cryptography • Essential mathematics for public key algorithms	• Class discussions • Class exercises
7	The RSA Cryptosystem	• Encryption and Decryption • Key generation • Speedup techniques for RSA • Implementation in software and hardware	• Class discussions • Class exercises • Lab exercises
8	Key management and public key cryptosystems	• Key management • Diffie-Hellman Key exchange • Elliptic curve arithmetic • Elliptic Curve cryptography	• Class discussions • Class exercises
9	CAT 2		
10-11	Message authentication and hash functions	• Authentication requirements and functions • Hash functions • HMAC • CMAC	• Class discussions • Class exercises • Lab exercise
12	Digital Signature and Authentication Protocols	• Authentication protocols • Digital signatures • Digital Signature Standards: RSA, DSA	• Class discussions • Class exercises • Lab exercise

5. Mode of Delivery

1. Lectures and tutorials
2. Lab exercises
3. Summary notes. Slides will be uploaded to the e-learning platforms
4. Discussions done in groups

6. Course Text

1. Cryptography and Network Security: Principles and Practice, Stallings, William, Pearson/, Prentice Hall, 2005 – 0131873164
2. **Understanding Cryptography, Christof Paar and Jan Elzi. Springer, 2010**
3. Security in Computing, Pfleeger, C. & Pfleeger, S., Prentice Hall, 2006, 0132390779
4. Introduction to Cryptography with Coding Theory, Prentice Hall, 2005, Trappe and Washington

7. Course Assessment

- Examination 60%
- CAT 1 5% Sit-in
- CAT 2 8% Practical
- Labs 10%
- Group work 5%
- Class Project 12%
- Total 100%

8. Assignments and/or Course Work

1. Plagiarism is a serious offence. If detected in any form in course work and assignments, the following will apply:
 - a) In partial or non-serious cases (such as not citing whole word-for-word quotes), half the total possible marks of the assignment are duly struck off.
 - b) In serious cases (such as whole duplication of a paper), a zero policy will apply i.e., all offending assignments will be awarded a mark of zero.

Note: The level of seriousness referred to above is at the discretion of the lecturer. Appeals are certainly possible through the relevant channels

2. Notwithstanding the above, collaboration in course work is certainly encouraged as this promotes team spirit and group synergy as long provided originality is preserved.

9. **Class attendance**

- At least 2/3 of the classes
- Problem with class attendance records to be addressed way before exam.

10. **Websites**

The course online resources and other materials will be available via the Strathmore University E-Learning System [SUES](#).

11. **Communication Channels**

- E-mail
- SUES
- Module Leader